

CourtFlow - Player-facing metrics and equations

Scope: metrics derived from player tracks only (detection + tracking + homography). Each track row has court coordinates (x_court, y_court) after calibration. Padel court constants W = 10 m, H = 20 m for path length in meters (Tier 1 / Tier 2). No ball model is required for these equations.

Notation

1) Distance (m), distance_m

$D_p = \sum_{i=2..N_p} L_i$ (skip if endpoint missing).

2) Tracked time (s), active_duration_s

3) Average speed (km/h), avg_speed_kmh

4) Fastest burst (km/h), max_speed_kmh

5) Sprint count, sprint_count

6) Side-to-side movement (%), lateral_movement_pct

7) Hard accelerations / decelerations

8) Time at net (%), net_pct

9) Time deep (%), baseline_pct

10) Court zones (0-5), zone_coverage_pct[z]

11) Rank (1-4), rank_by_distance

12) Share of running (%), share_of_distance_pct

13) Heatmap (movement map)

Data inputs

p = player
and i. dt
duration (

$L_i = \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}$

$T_p = t_{last} - t_{first}$

$v_{avg} = D_p / T_p$

$v_i = L_i / dt_i$

Segment

As in
 $\sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}$
normalize

From seg
m/s^2. de

net_pct =

baseline_

Each sam

Sort play

share_p =

2D histog

tracks.js
(x_pixel, y